6 In an X-ray tube, electrons are produced from a filament heated by an electric current as shown in Fig. 6.1. A large accelerating potential difference is set up between the filament and the target material. The electrons are accelerated from the filament and hit the target material to emit X-ray photons.

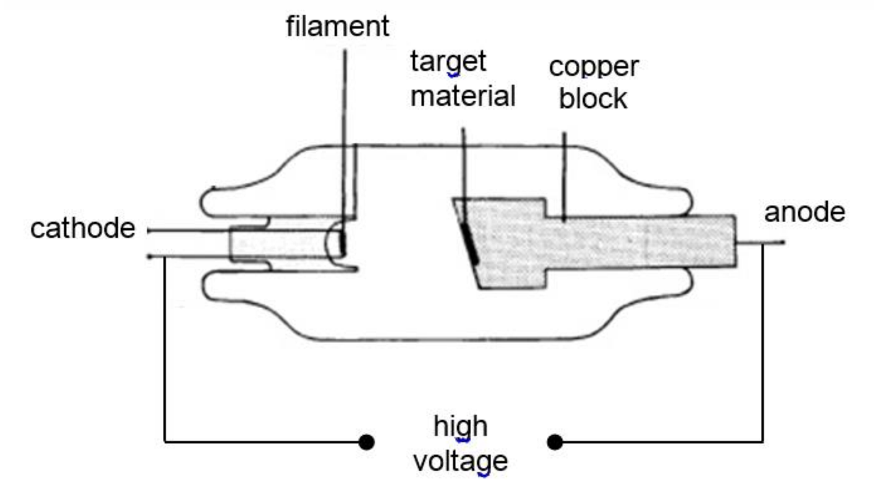


Fig. 6.1

A graph of intensity against wavelength of the emitted radiation is plotted as shown in Fig. 6.2 when the X-ray tube is operated at a voltage of 50 kV.

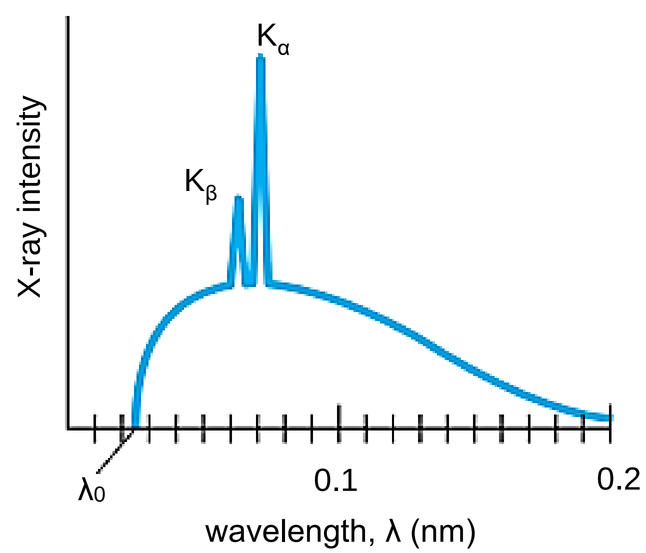


Fig. 6.2

(a) Explain why there is a minimum wavelength λ_0 for the emitted X-rays.

....

.... [2]

When an **accelerated electron from the cathode loses all of its kinetic energy (KE) in a single head-on collision with the target atom at the anode**, **1**

the **KE lost is converted into an x-ray photon of maximum energy**, **hence minimum wavelength**. **1**

(b) Show that λ_0 equals to 0.024 nm.

[1]

From (a),
Loss in EPE as electron moves from cathode to anode = Maximum KE gained just before it hits the anode = Maximum energy of photon emitted

So,
$\boldsymbol{eV_0 = hc/\lambda_0}$
$\Rightarrow \boldsymbol{\lambda_0 = hc/eV_0}$
$\Rightarrow \boldsymbol{\lambda_0 = (6.63 \times 10^{-34})(3.00 \times 10^{8}) / (1.60 \times 10^{-19})(50 \times 10^{3})}$ **1**
$\Rightarrow \lambda_0 = 2.48625 \times 10^{-11} = 0.0248625 \times 10^{-9}\ \text{m} \approx 0.024\ \text{nm}$

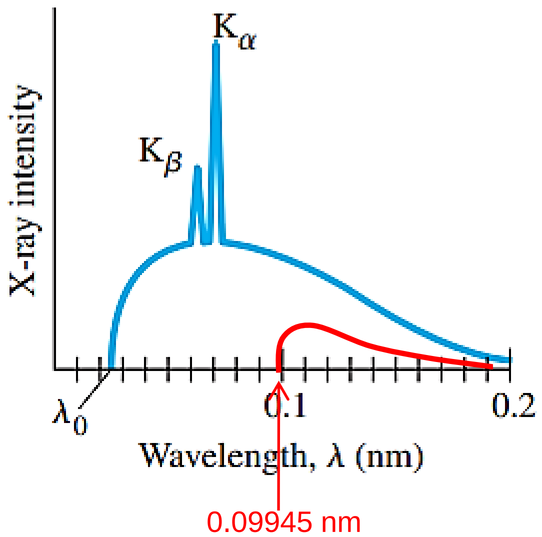
(c) Sketch on Fig. 6.2 a graph to show the intensity variation with wavelength if the accelerating potential difference is reduced to one-quarter of its original value. [2]

Graph of **similar shape**, **below** the original graph (intensity lowered at all wavelengths), **1**

$\lambda_0 = hc/eV_0$
Since h,c,e are constants, thus $\boldsymbol{\lambda_0}$ **is inversely proportional to** $\boldsymbol{V_0}$.

Thus **if** $\boldsymbol{V_0}$ **is one-quartered, $\boldsymbol{\lambda_0}$ will be increased by 4 times!**

New $\boldsymbol{\lambda_0}$ = 4 x 0.0248625 nm = 0.09945 nm ≈ 0.1 nm AND without any characteristic lines. **1**



7	An induced nuclear fission reaction may be represented by the equation